

Topic: Performance Measures of Computer Architecture

Q. How is the performance of a computer system measured? Discuss CPU Execution Time, Clock Rate, and the various factors that affect computer performance.

> **Definition to Remember**

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CPU Performance Measures

The actual time the CPU spends computing a specific task (excludes I/O wait time).

Form

2. Clock Rate and Clock Cycle

- **Clock Cycle (Period):** The fundamental unit of time in a computer (e.g., 1 ns), generated by a hardware oscillator.

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3. Other Key Performance Metrics

- **CPI (Cycles Per Instruction):** Average number of clock cycles needed to execute one instruction.

- Exe

4. Factors Affecting Computer Performance

| Factor | Impact on Performance |

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> **Must-Write Points (for 10 marks)**

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> **Quick Recall**

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Speed